

BIOMECHANICS (EPHE 341)

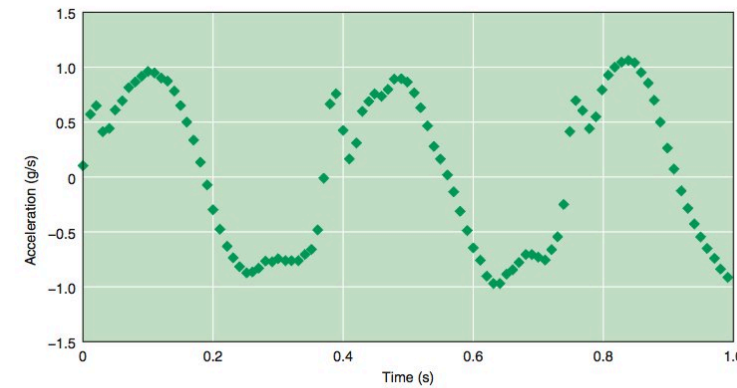
Signals

Separating the true signal from the noise.

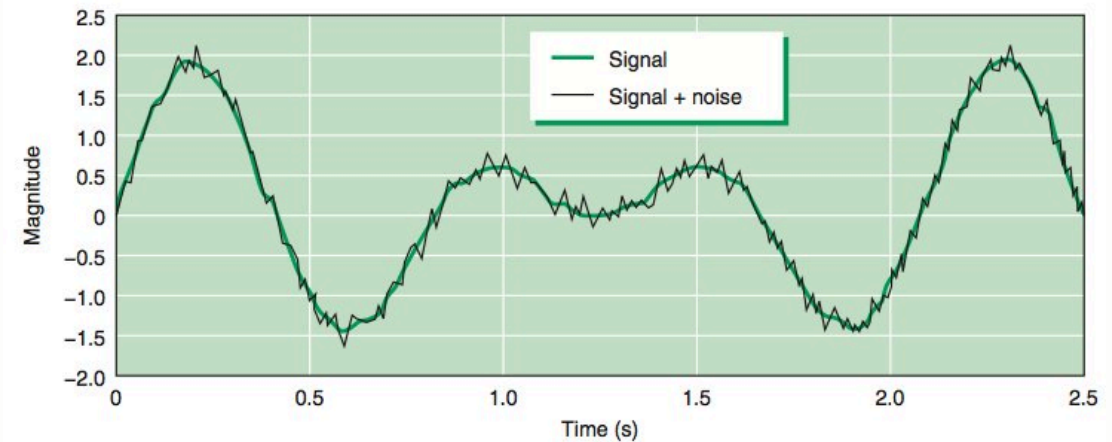
In biomechanics we are dealing with values from sensors that have limitations based on

- Sampling
- Sensor error
- Environmental error

We need to be able to separate the **true signal** from the **noise**



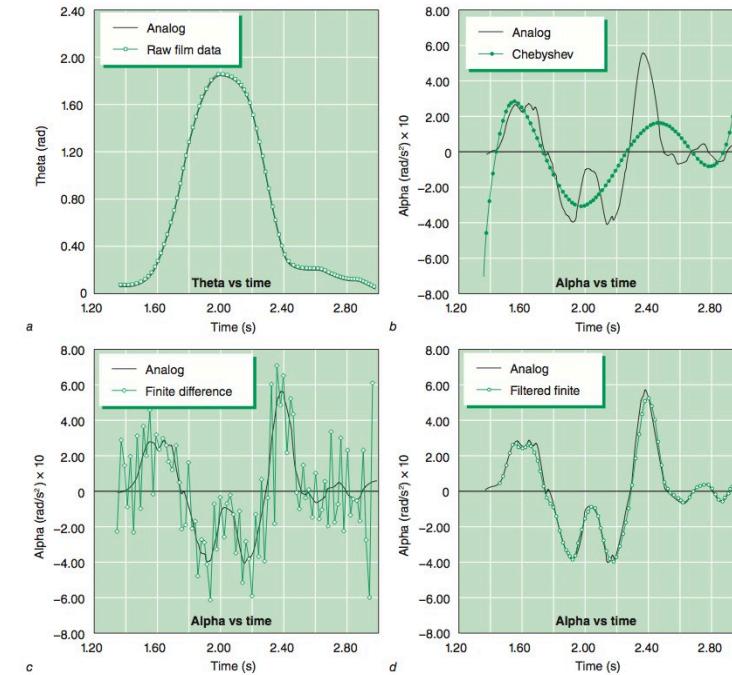
▲ **Figure 12.1** The digitized discrete representation of the acceleration of the head while a subject is running. The signal was sampled at 100 Hz (100 samples per second).



We can minimize the noise by **processing** the data.

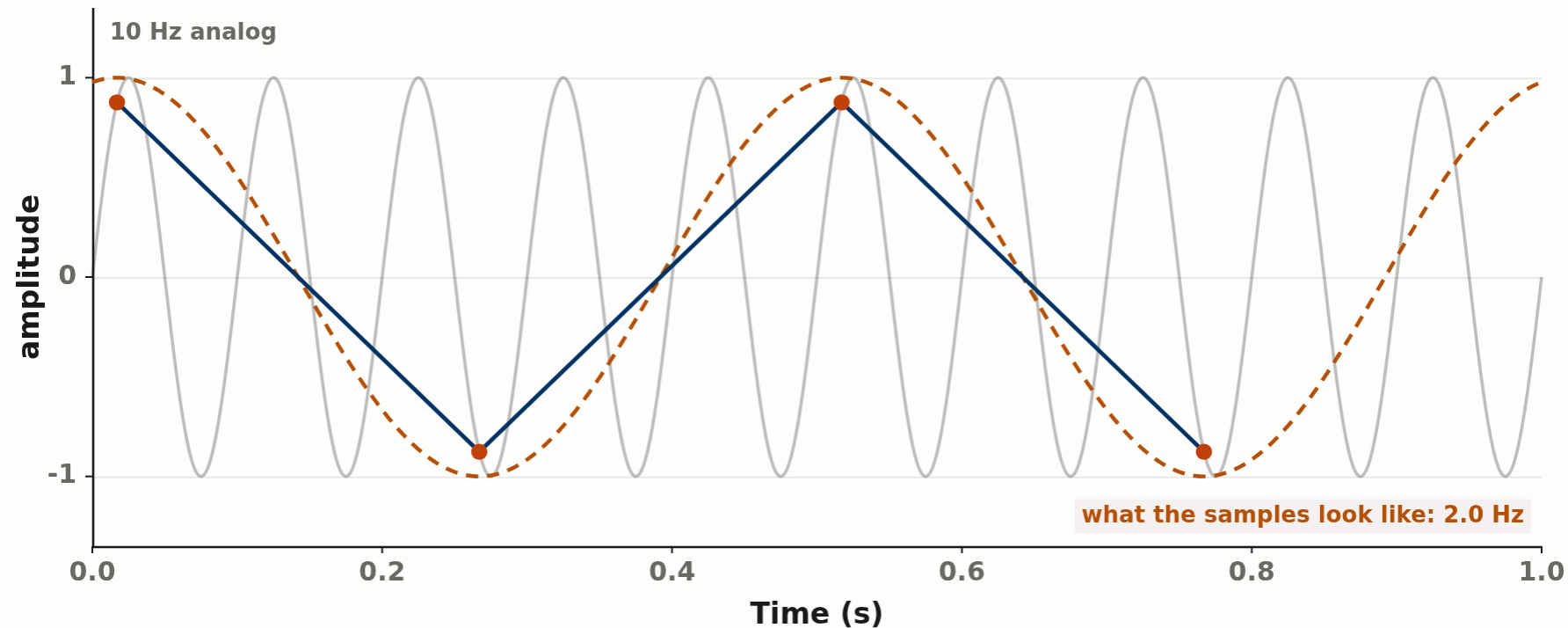
- Sampling properly
- Smoothing data
- Filtering data
- Fitting data

To inform our processing choices we need to understand some basic characteristics of signals and the transformations that we intend to perform on them.



Nyquist–Shannon Sampling Theorem

In sampling an analog signal, the goal is to represent the original signal as accurately as possible with as few discrete points as possible. **How high must the sampling frequency be for an analog signal to be accurately represented by discrete data points?**

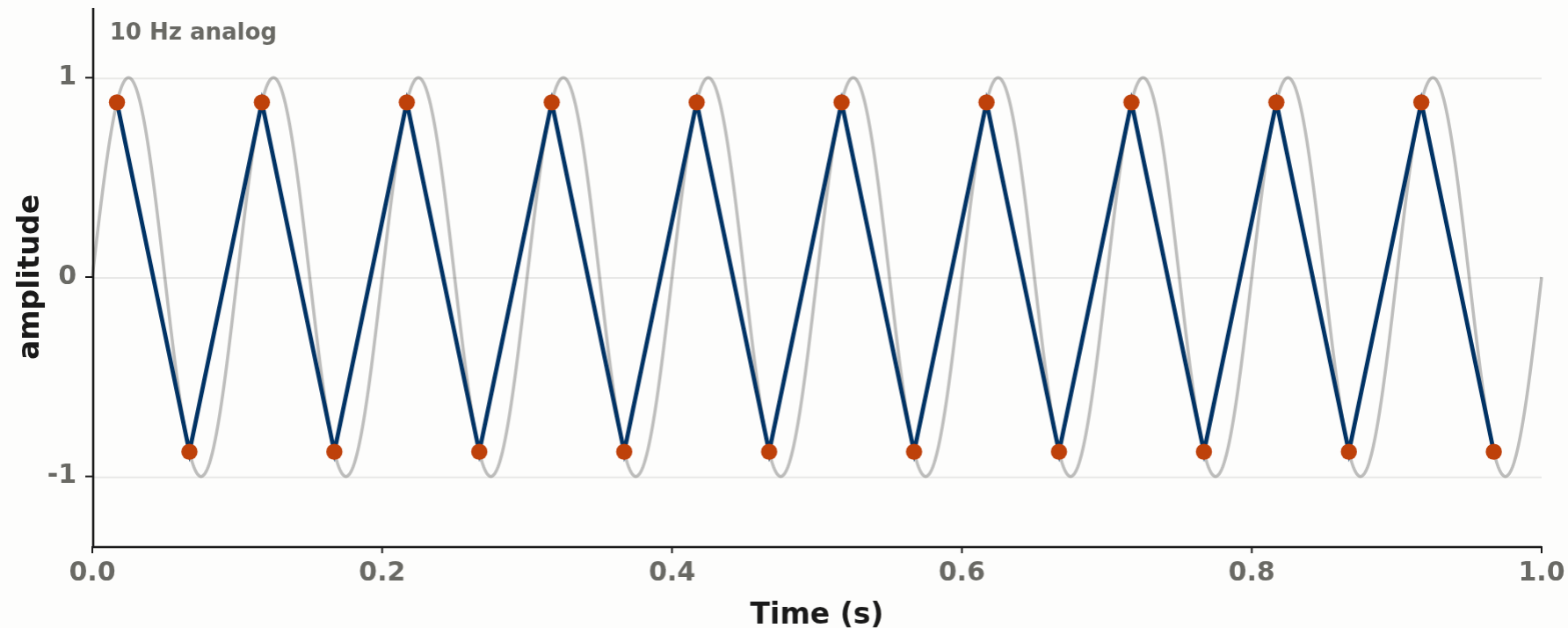


sampling at **4 Hz** · 4 samples in a second · $0.4\times$ the signal frequency

below the Nyquist frequency — **aliased**. The dashed line is a **different** sine wave that fits every sample exactly as well. Nothing in the data can tell them apart.

Nyquist–Shannon Sampling Theorem

For a digital signal to represent an analog signal perfectly, the sampling frequency must be **at least twice** the analog signal's highest frequency.



sampling at **20 Hz** · 20 samples in a second · 2.0× the signal frequency
above Nyquist — the shape is right but sharp turns are rounded off. The Nyquist frequency here is 20 Hz.

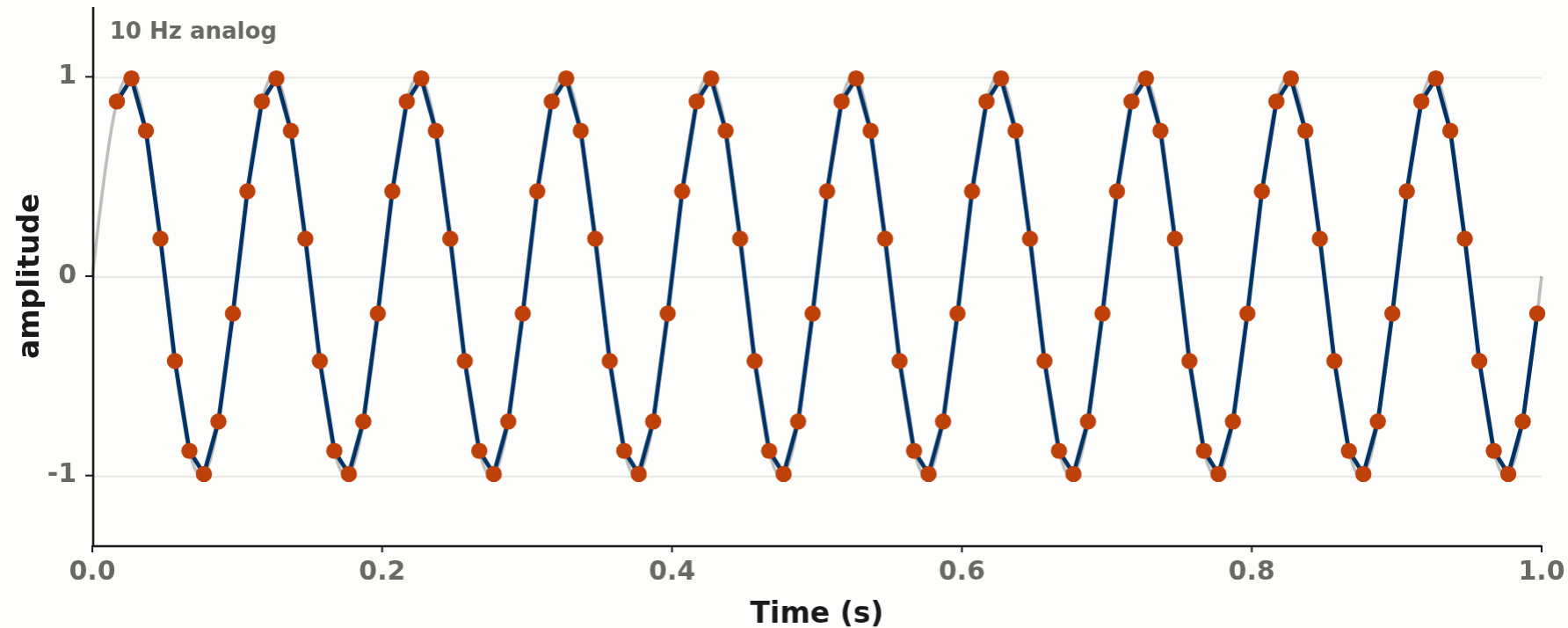
Sampling
frequency



20 Hz

Nyquist–Shannon Sampling Theorem

To ensure that sharp direction changes in the signal are not missed, sampling frequencies of **5–10 times** the Nyquist Frequency are often used.



sampling at **100 Hz** · 99 samples in a second · 10.0× the signal frequency
5–10× the signal frequency — direction changes are caught. The Nyquist frequency here is 20 Hz.

Sampling
frequency

100 Hz

Following data collection, “raw” data often contains noise.

Processing data post collection can improve the **signal to noise (S/N) ratio**.

Smoothing, filtering and curve fitting can improve S/N ratio.

Processing is necessary **before** calculations are performed.



Filtering, smoothing and fitting can improve S/N using different approaches.

Filtering:

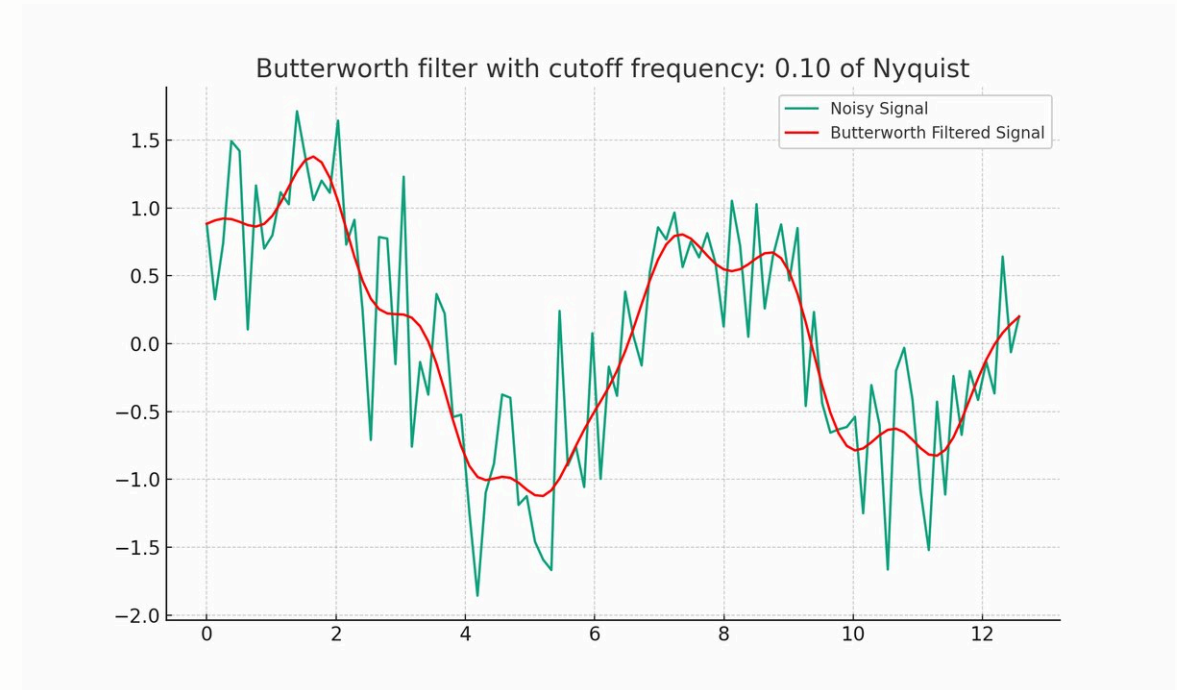
involves altering or removing certain frequency components from a signal.

Smoothing:

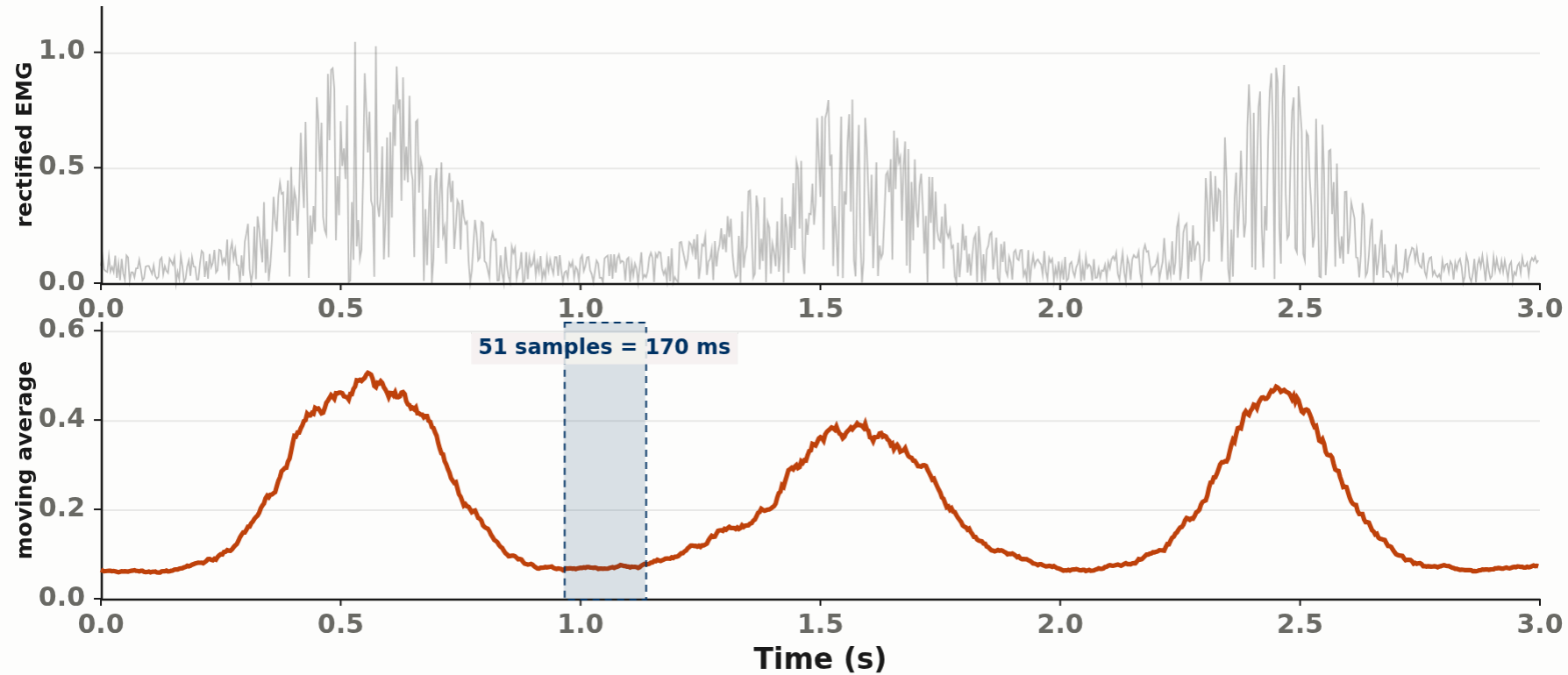
involves averaging or aggregating data points to reduce short-term fluctuations and highlight longer-term trends or cycles.

Curve Fitting:

to find a mathematical function that best represents a dataset. This function may be a simple linear equation, a complex polynomial, or any other



Moving average (over a window of samples). Simple averaging gets rid of noisy spikes.



window of **51** samples · 170 ms at this sample rate

A wider window is a smoother line — and a later, lower, blunter one. Watch the height of the bursts fall and their edges soften as the window grows past the length of the burst itself. Simple averaging gets rid of noisy spikes, but it cannot tell a spike from a fast real change.

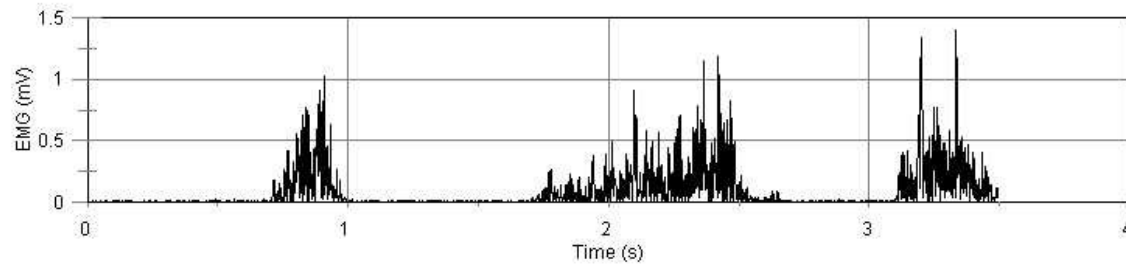
Window



51 pts

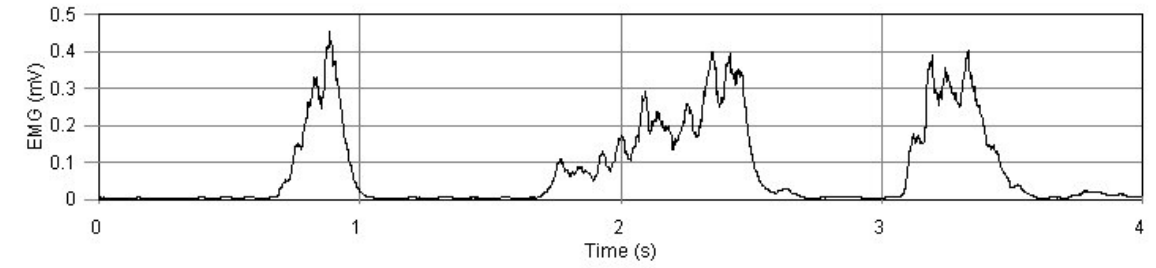
Smoothing

Full-wave rectified EMG



EMG (1010 Hz sampling rate)

Averaged EMG (window of 51)

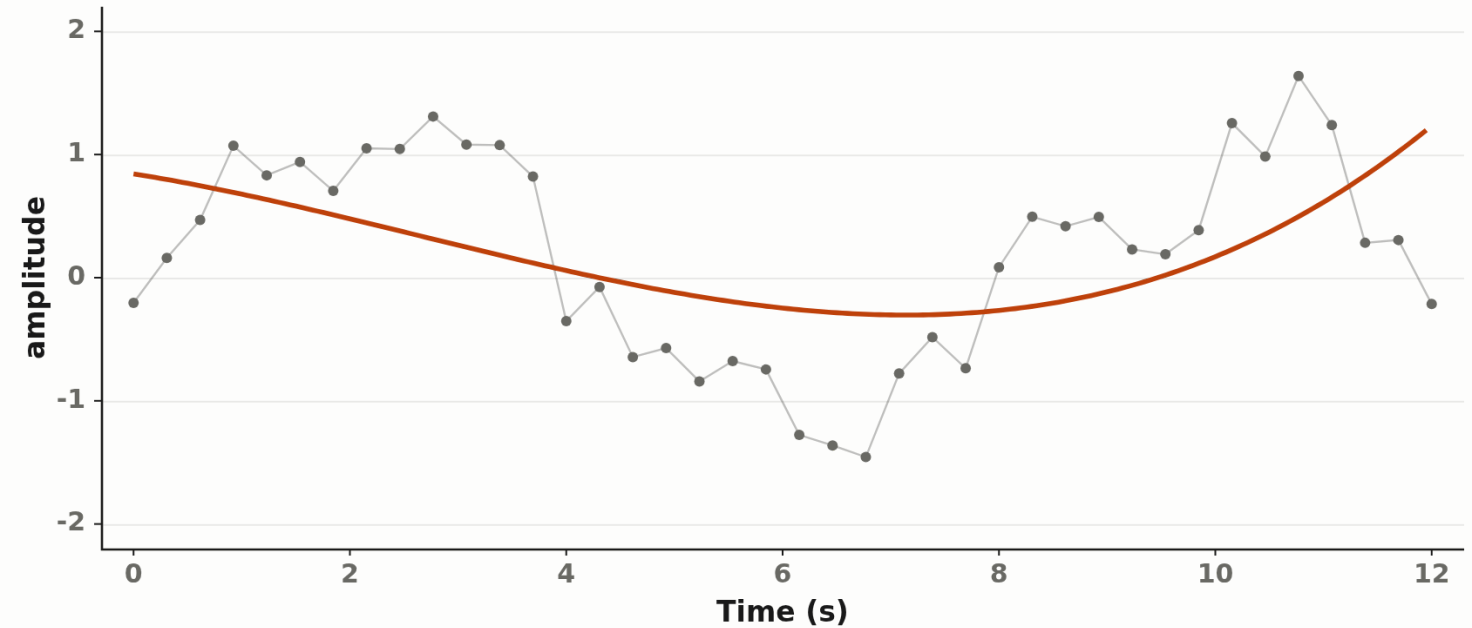


averaged EMG (moving average, 51 points)

Curve fitting

Fits a **best-fitting function** to the data.

- Polynomials
- Splines
- linear



polynomial of degree 3 · residual RMS **0.690**

Raise the order and the residual always falls — that is arithmetic, not insight. Somewhere past a handful of terms the curve stops describing the signal and starts describing this particular sample of noise.

linear

polynomial

spline

Polynomial order

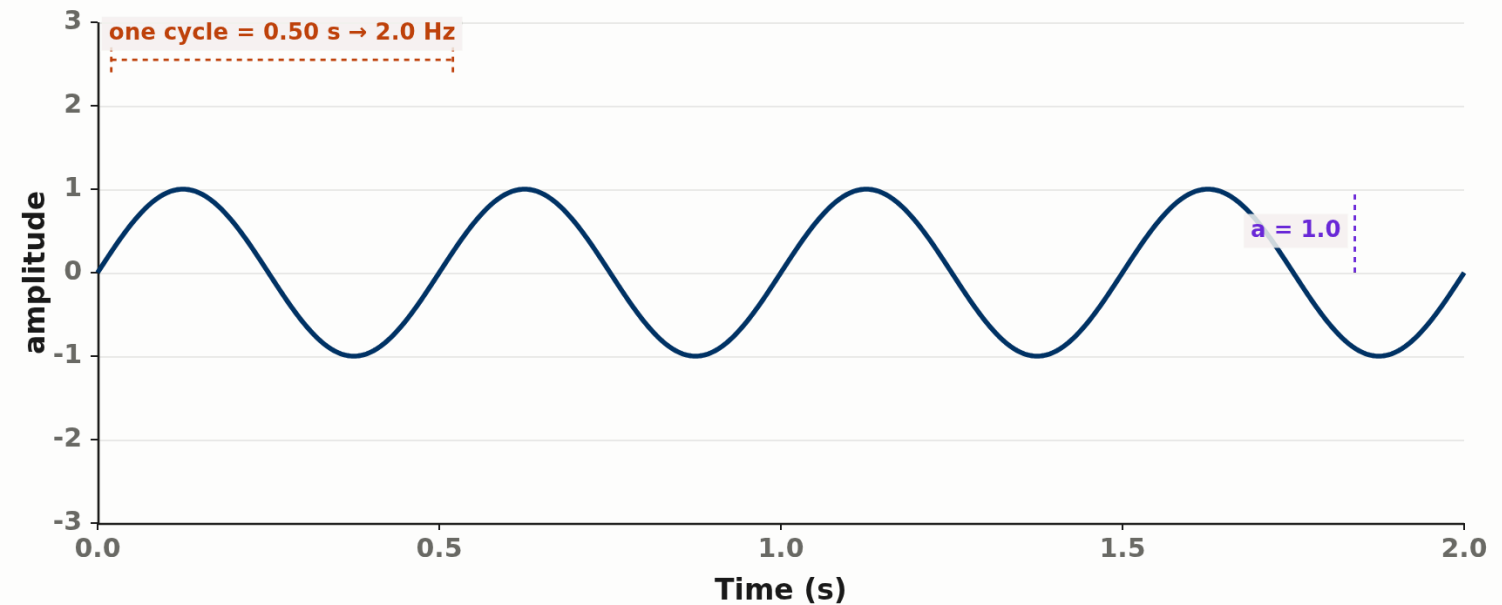


3

Components of a signal

Filtering – a process that removes some unwanted components or features from a signal. Before we discuss filtering we need to talk about **signal components**.

- **frequency:** the number of repeating cycles per unit time
- **amplitude:** the magnitude of change
- **offset:** the distance of the mean amplitude from zero
- **Phase Shift:** a timing offset



$$y(t) = 0.0 + 1.0 \cdot \sin(2\pi \cdot 2.0 \cdot t + 0^\circ)$$

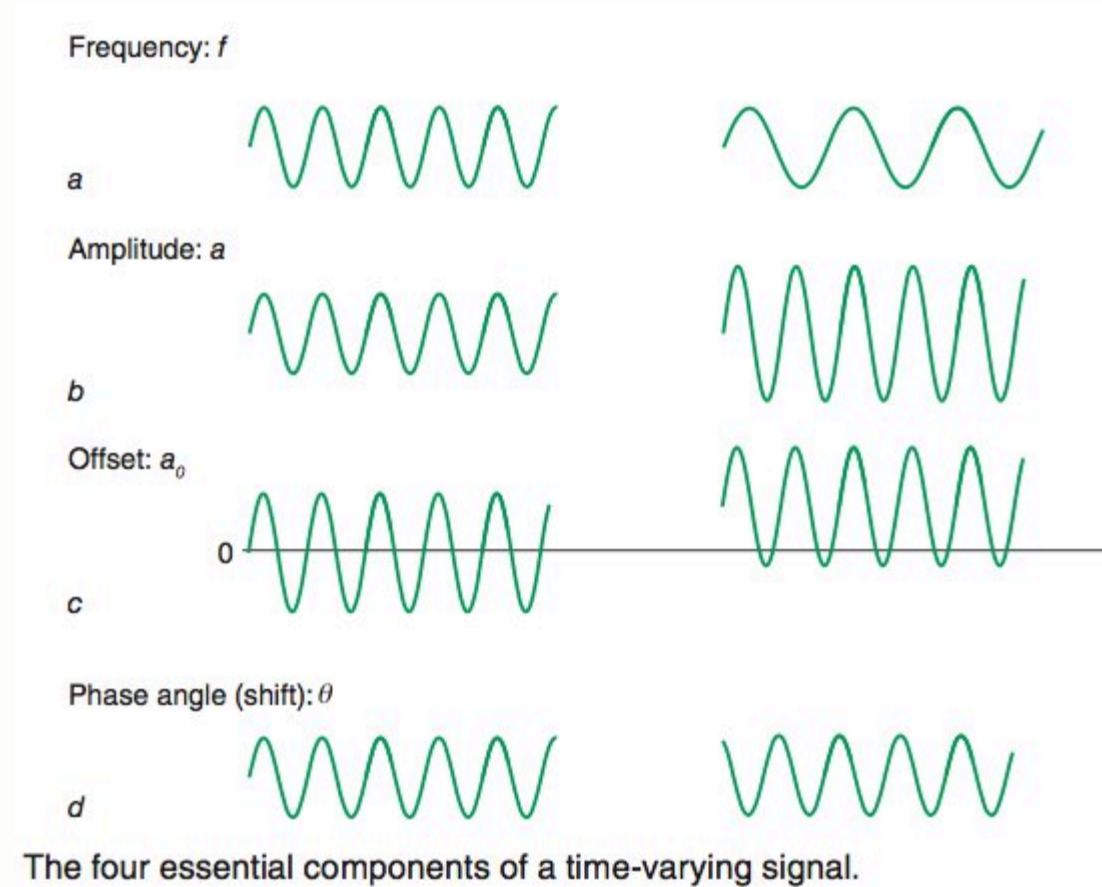
The dashed grey wave is the starting signal. Frequency stretches it along time, amplitude stretches it up and down, offset lifts the whole thing off zero, and the phase shift slides it sideways without changing its shape.

Frequency

Amplitude

Offset

Components of a signal



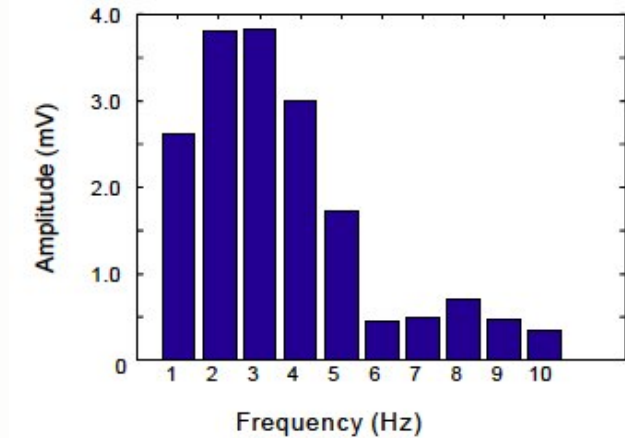
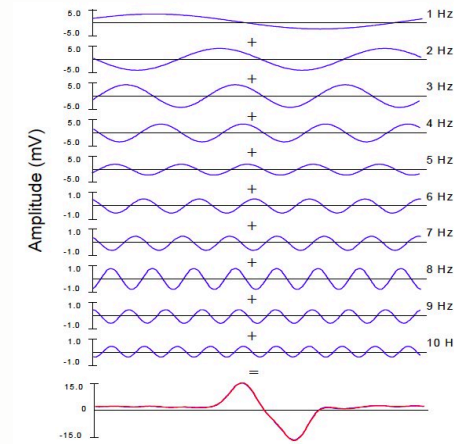
The four essential components of a time-varying signal.

Signals

Signals can be **decomposed and reconstructed** from a number of basic functions (sine waves).

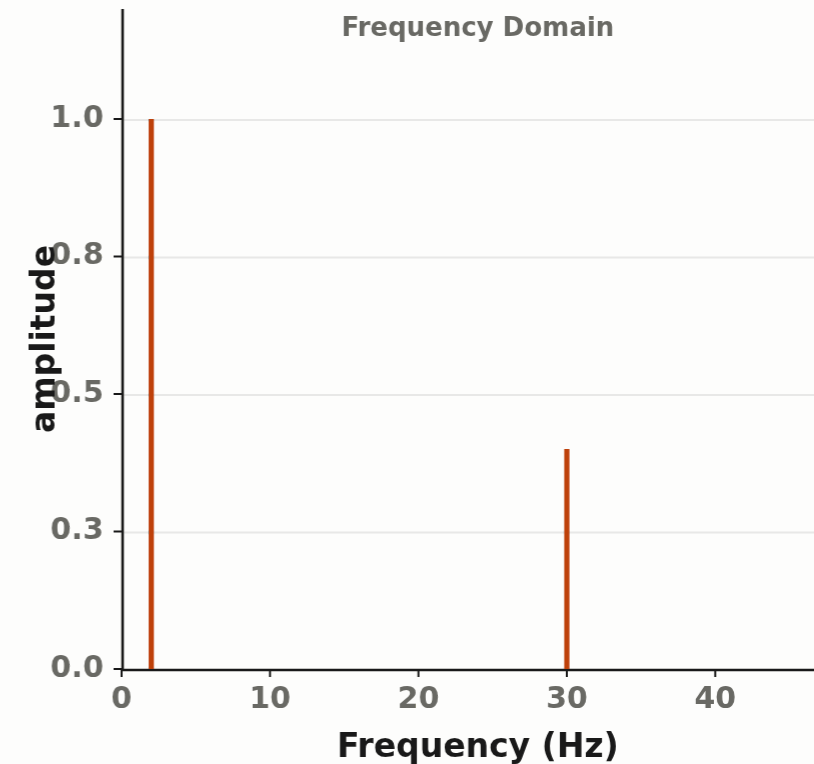
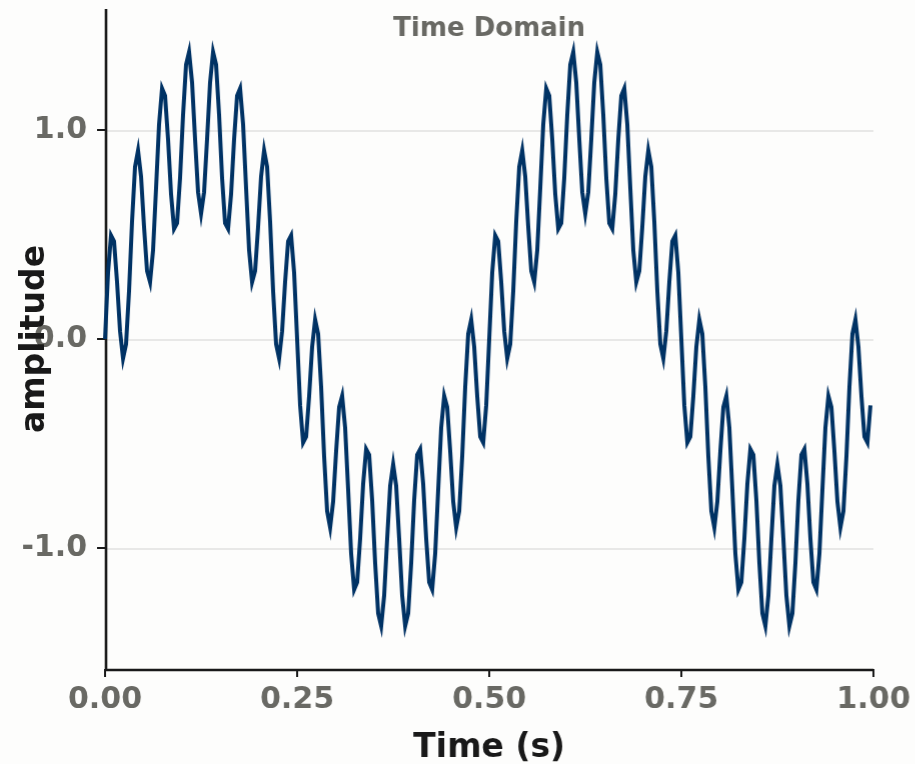
Once a signal is decomposed it can be represented in a frequency graph of the basic sine waves.

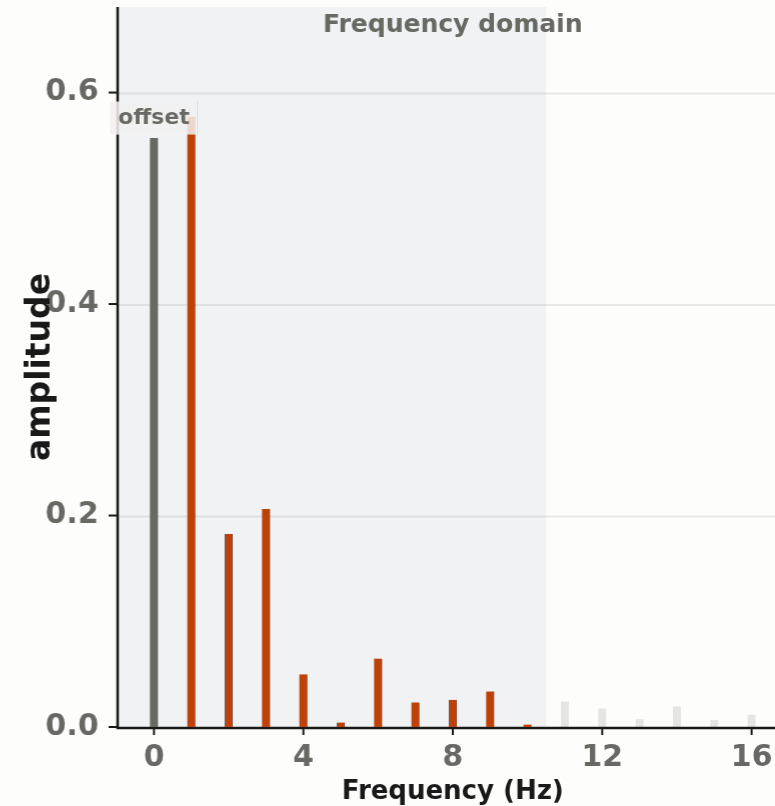
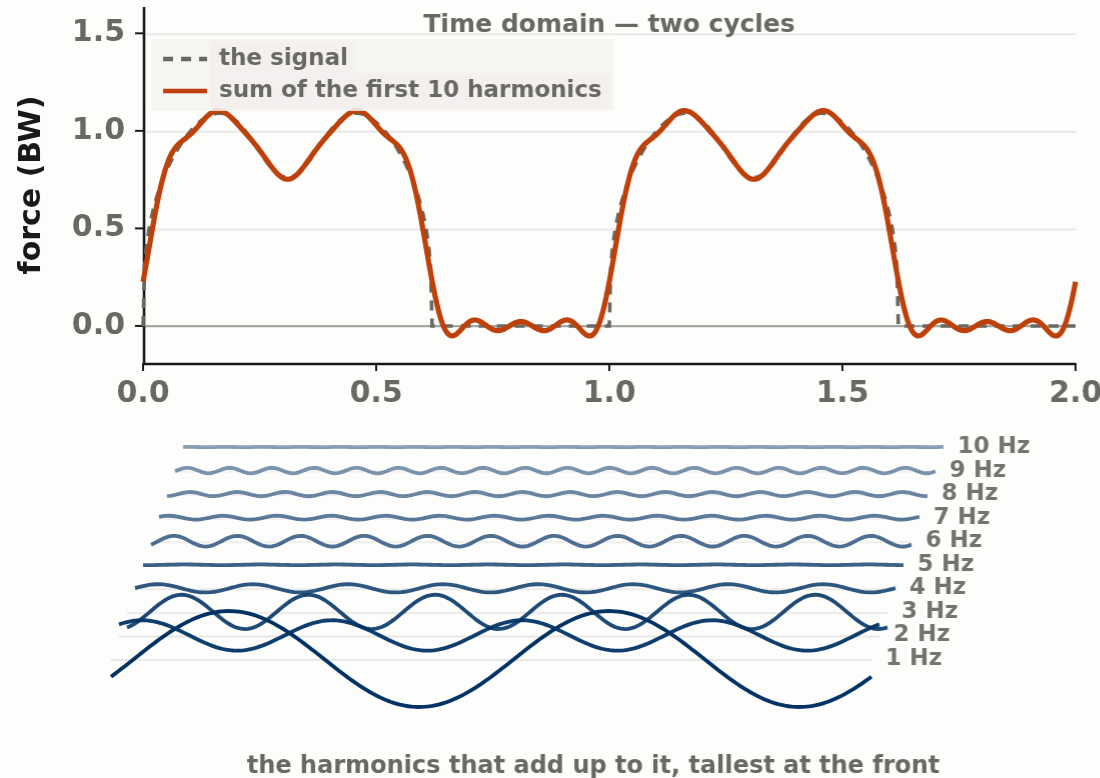
Time Domain → Frequency Domain



Frequency domain

A time-domain signal is converted to a frequency-domain signal with a mathematical technique called a **Fourier Transform**.





$$x(t) \approx 0.56 - 0.21 \cos(2\pi \cdot 1t) + 0.54 \sin(2\pi \cdot 1t) + 0.13 \cos(2\pi \cdot 2t) + 0.13 \sin(2\pi \cdot 2t) - 0.19 \cos(2\pi \cdot 3t) + 0.09 \sin(2\pi \cdot 3t) \dots$$

10 harmonics — every whole multiple of the 1 Hz fundamental up to **10 Hz**. Residual RMS **0.037**. Six to eight harmonics carry almost all of a walking GRF — which is where the usual 6 Hz cutoff comes from.

Walking GRF

Running GRF

Square wave

one wave per harmonic

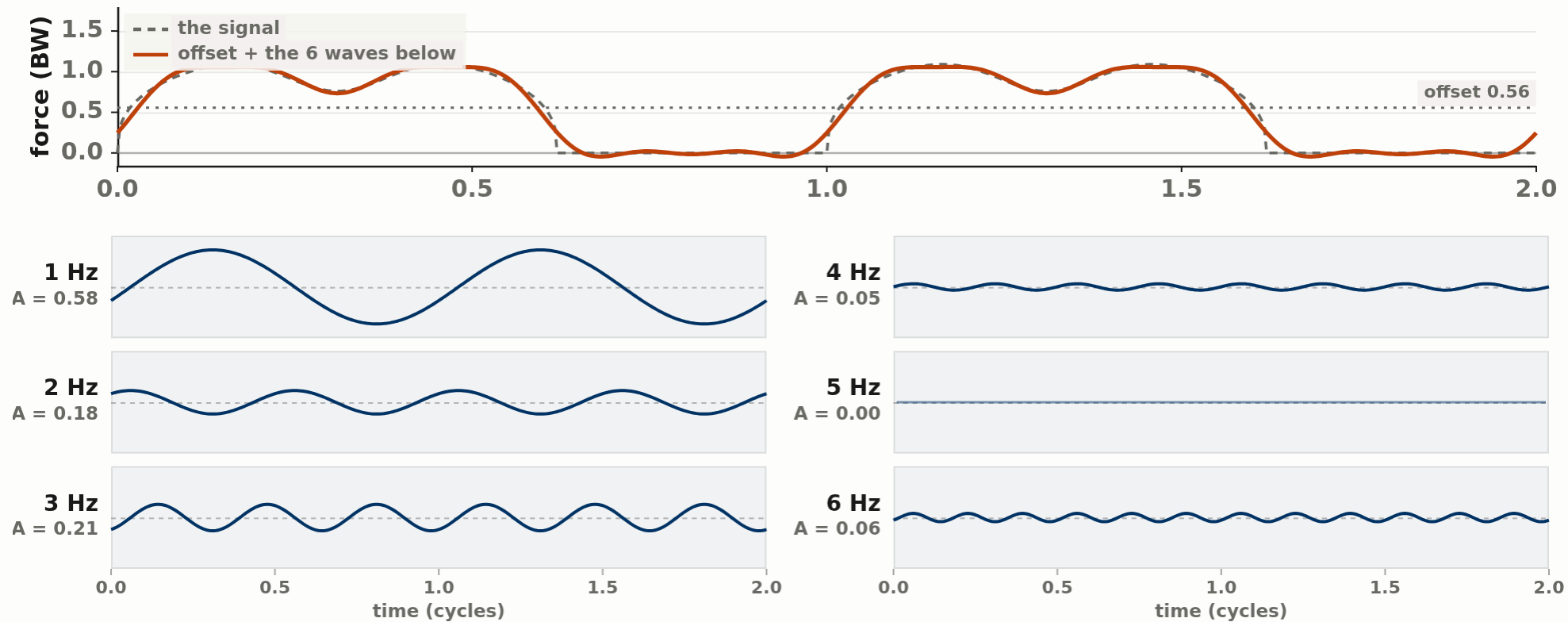
sin & cos parts

► Build up

Harmonics kept

10 of 16

The components, one graph each



Walking GRF = 0.56 offset + the 6 waves below, added together

One flat graph per harmonic. One shared amplitude scale, so a term drawn flat really is contributing almost nothing.

Walking GRF

Running GRF

Square wave

one wave each

sin & cos

one scale

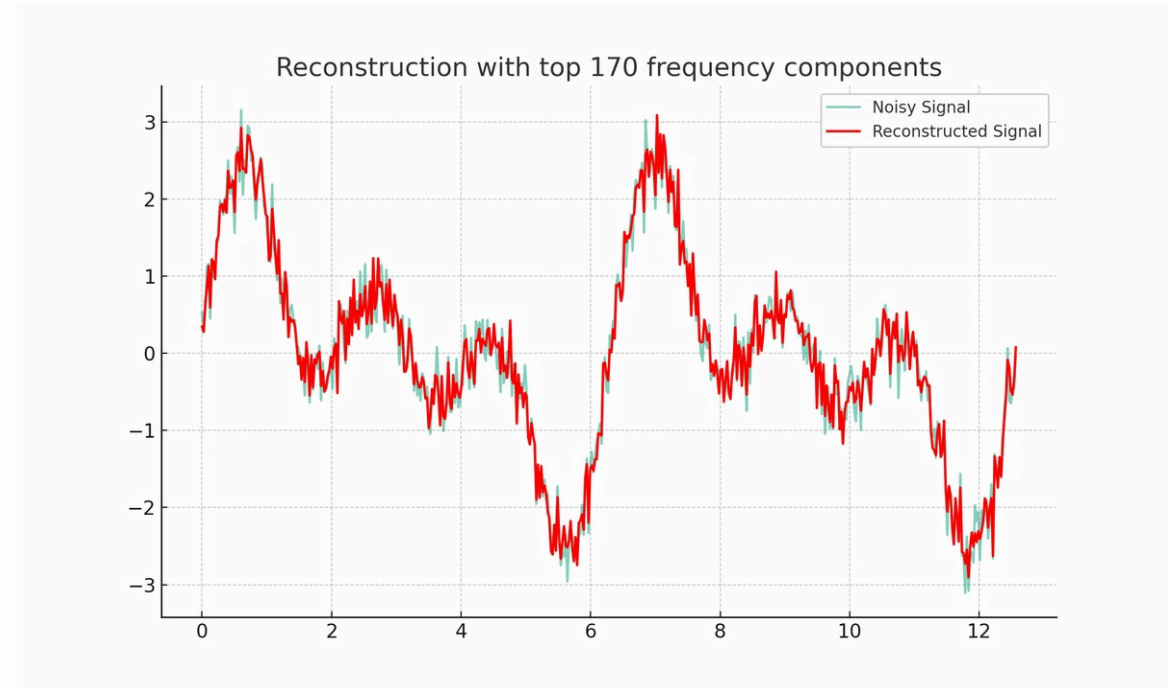
each to fit

Digital filtering

Is a way of reducing the frequency content of the signal so that it just obtains the signal you are interested in without the noise.

A common use of Fourier transforms is to find the frequency components of a time-domain signal that is mixed with noise.

Because noise is often of a different frequency than the intended signal, a filter can be used to remove the noise (ie: parts of the signal that are a certain frequency).



Types:

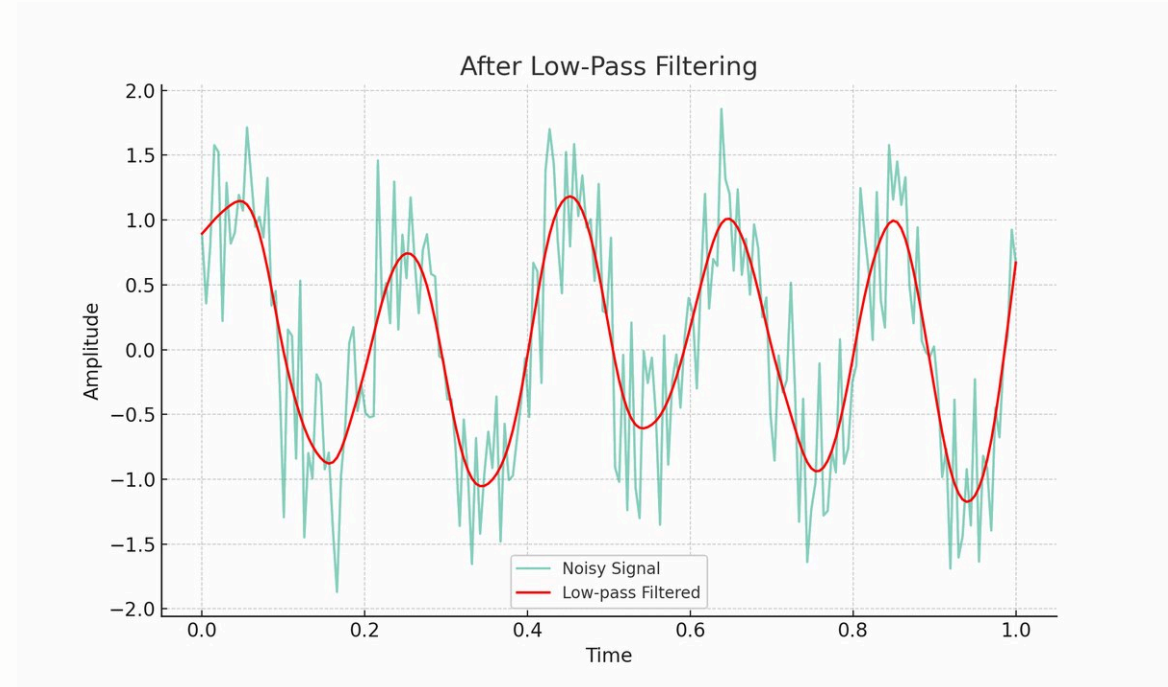
low-pass, high-pass, band-pass, band stop and notch (single or small band-stop filter, useful for AC interference)

Designs:

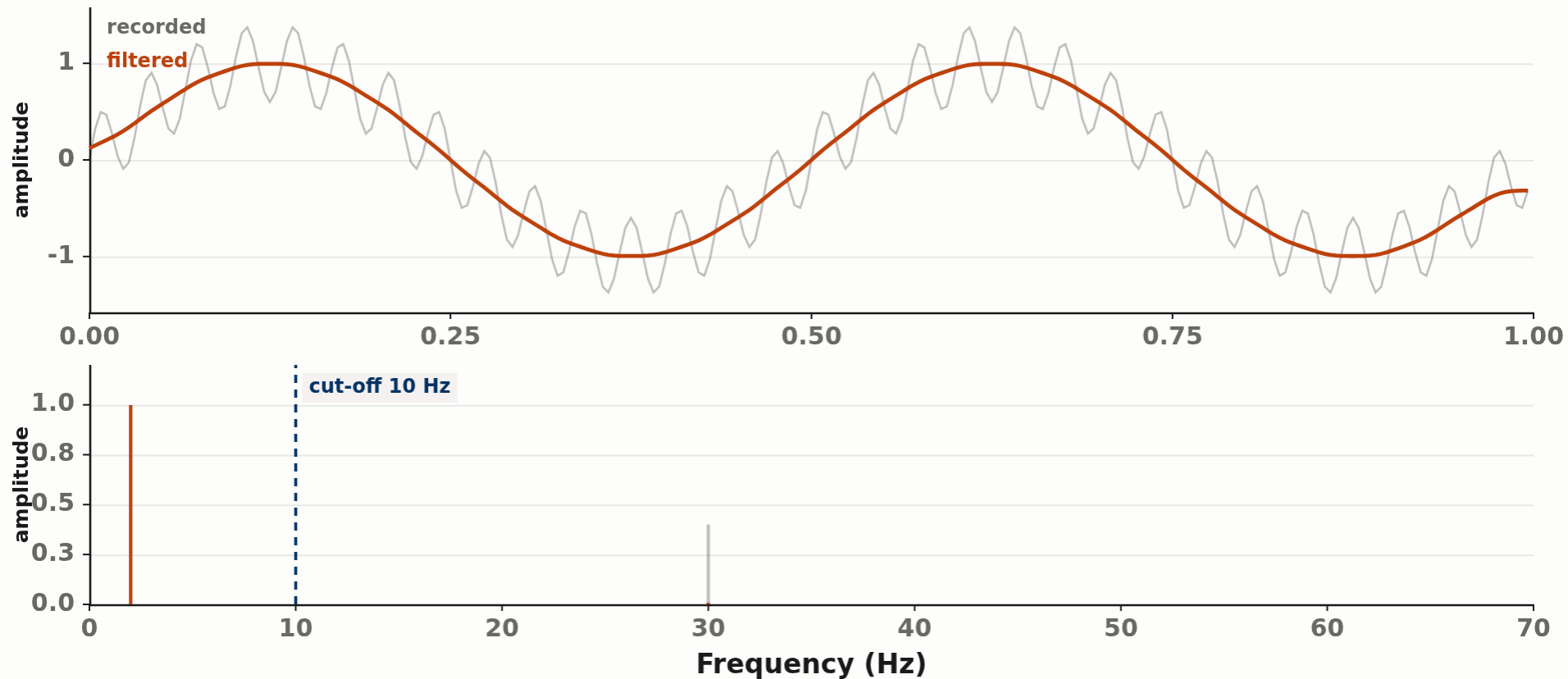
Butterworth, critically-damped, Chebyshev etc. (sharper cutoffs), Generalized Cross-validation (GCV also called Woltring filter, 1986)

Problems:

Phase distortion (usually phase-lags occur)



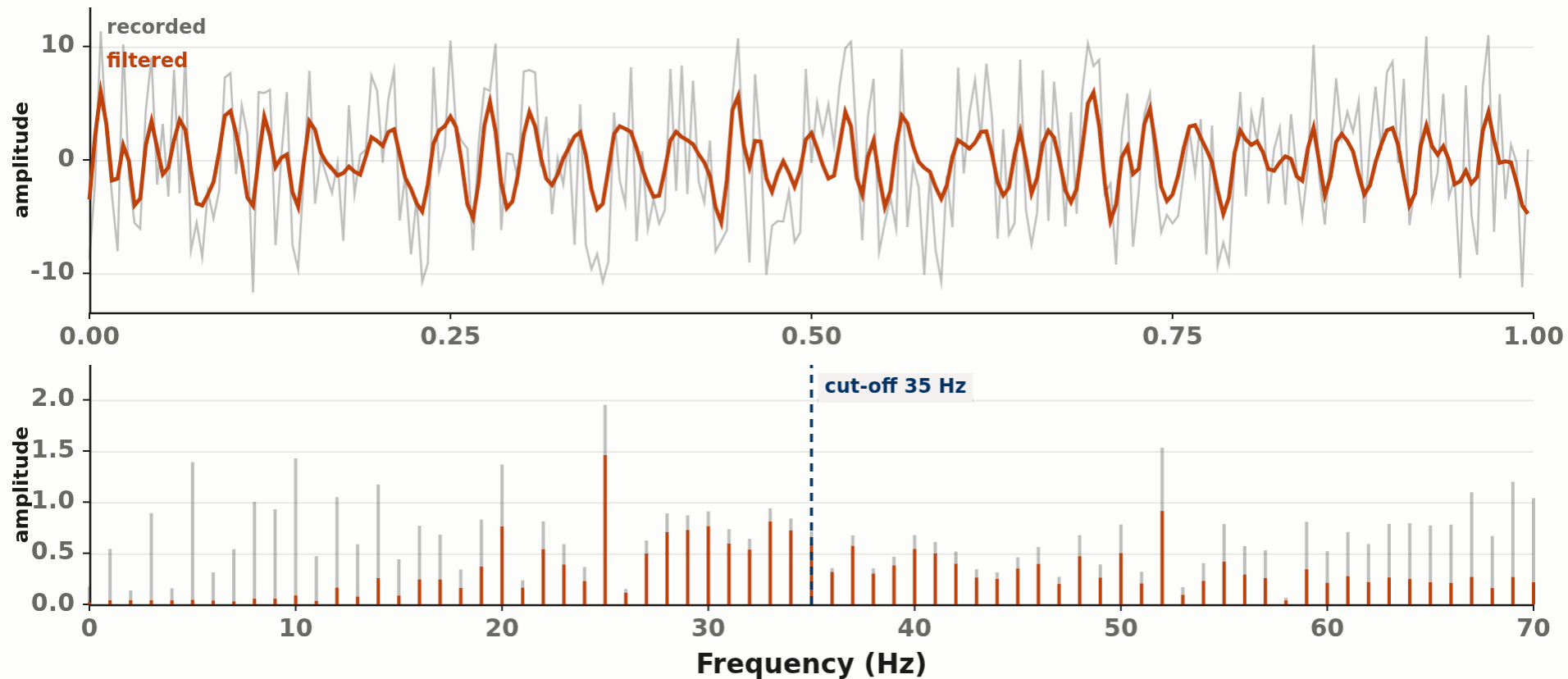
Let's say that we wanted to isolate different parts of the signal — we could use different filters to do that. A **low pass filter** maintains low frequencies; a **high pass filter** maintains high frequencies.



low-pass at 10 Hz · 93% of the original amplitude survives

The grey spikes are what was recorded, the red ones what is left. A filter does not make a signal cleaner — it decides which frequencies to keep, and everything at those frequencies, signal or noise, is kept with them. This one is run forwards and then backwards, which removes the phase lag that a single pass would leave behind.

25 Hz + 50 Hz, buried in random noise **ten times larger** than the sine waves themselves — and pulled back out again.



band-pass at **35 Hz** · **42%** of the original amplitude survives

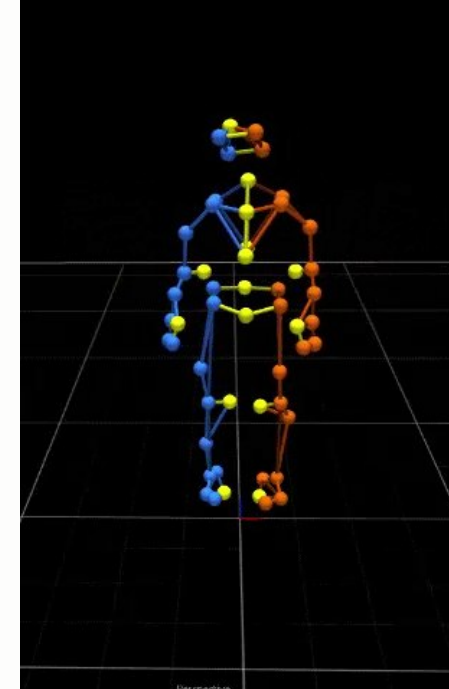
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Digital filtering

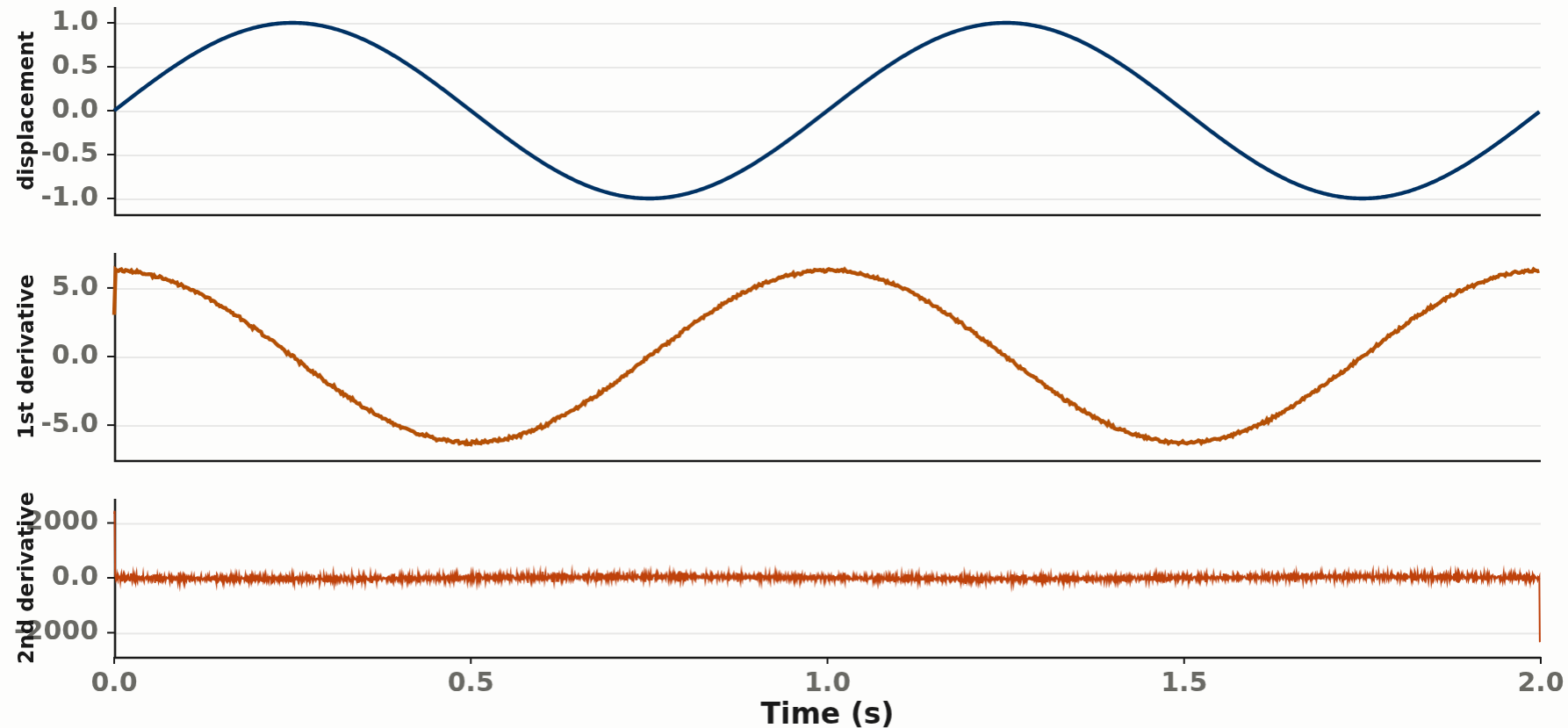
The most common source of noise in human motion capture data are small errors in the position of the biomechanical markers during the digitization process.

This noise is typically **low amplitude and high frequency**; therefore, a low-pass filter is often used to smooth kinematic data.

The aim of filtering is to arrive at the highest **signal : noise** ratio



Noise in motion capture data is **amplified** each time the data are differentiated or integrated.



noise amplitude **1e-4** — a signal-to-noise ratio of **10000:1**

The noise is far too small to see in the top trace. Differentiating multiplies every frequency by its own frequency, so high-frequency noise grows and the 1 Hz signal does not — by the second derivative it is all you can see. This is why kinematic data is filtered **before** it is differentiated, not after.

Digital filtering

Filtering, smoothing and fitting can aid in signal **Event Detection**.

In many biomechanical analysis, signal processing algorithms are required.

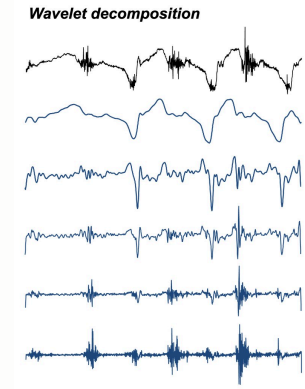
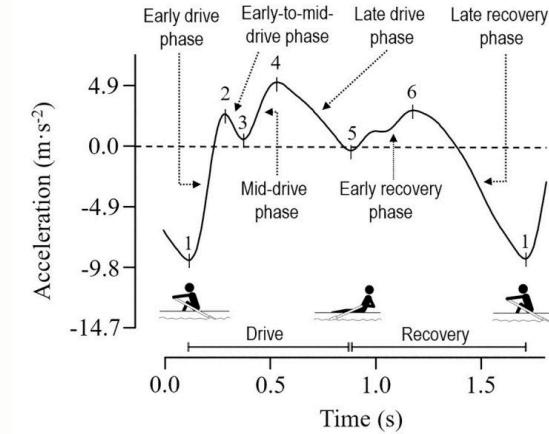


Figure 2. 5 level wavelet decomposition of hull acceleration signal.

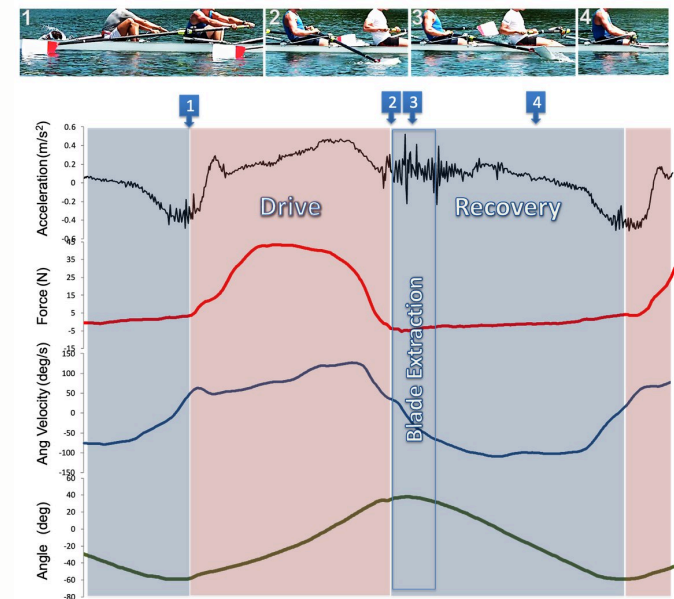
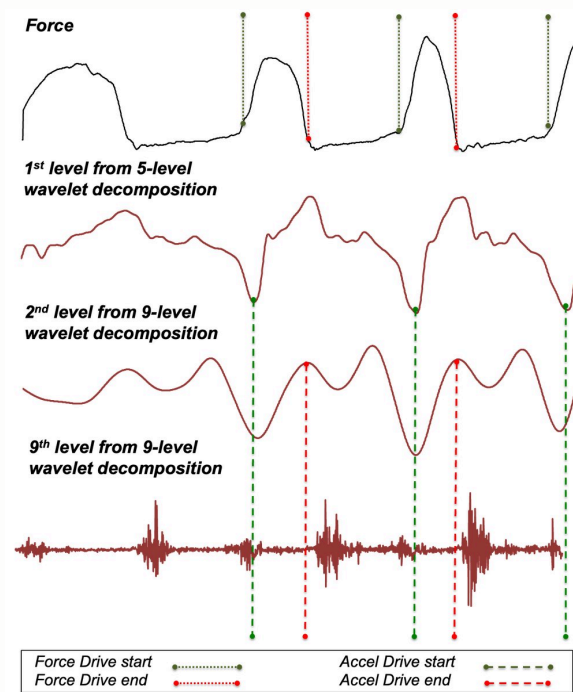


Figure 1. Rowing hull acceleration, force, oar angle and angular velocity data showing important phase features.

- It is important to know features of the signal we are collecting.
- Highest natural frequency in signal helps define the **minimum sampling rate**.
- Data processing can come in the form of **Filtering, smoothing and fitting**.
- It is important to get rid of noise **before and after** data transformation.